

Indian Statistical Institute
Second Semester Examinations: 2025-26

Course Name: M. Math, 2nd year

Subject Name : PDE

Maximum Marks: 40, Duration: Two hours

Date: 21.02.2026, 02:30 PM – 04:30 PM

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- You may use any results proved in class. Any other results require proof.
 - You will be awarded the full marks for the question if you justify using correct arguments that there is a mistake in the question.
 - All notation used below have same meaning as defined during the lectures.
 - Information you may use to assist your memory:

1. Poisson kernel for the upper-half space H_n^+ :

$$K(x, y) = \frac{a_n x_n}{\|x - y\|^n} \quad \forall x \in H_n^+, y \in \partial H_n^+.$$

2. Poisson kernel for the ball $B(0, r)$:

$$K(x, y) = \frac{b_n(r^2 - \|x\|^2)}{r\|x - y\|^n} \quad \forall x \in B(0, r), y \in \partial B(0, r).$$

- Unless otherwise stated, $V \subset \mathbb{R}^n$ is an open, C^1 , connected, bounded set such that

$$G_V(x, \cdot) \in C^2(\bar{V}, \mathbb{R}),$$

and $2 \leq n \in \mathbb{N}$.

1. For $z \in \mathbb{R}^n$, let

$$K_z = \text{Stab}_z(E_n)$$

denote the stabilizer of z in E_n , where E_n is the group of isometries of $\mathbb{R}^n$, equipped with the Euclidean metric, that is the group of affine maps preserving the Euclidean metric. Consider the following statements.

- (i) There exists $x \in \mathbb{R}^n$ and $r > 0$ such that $V = B(x, r)$.
- (ii) There exists $x \in \mathbb{R}^n$ such that for any $y \in V$, $y \neq x$, if $k(y) \in V$ for an element $k \in K_x$, then it holds

$$G_V(x, k(y)) = G_V(x, y).$$

Show that

- (a) [3 marks] (i) implies (ii).
 (b) [7 marks] (ii) implies (i).

2. [5 marks] Let $2 \leq n \in \mathbb{N}$. Let $V \subset \mathbb{R}^n$ be an open, C^1 , connected, bounded set. Let $U \subset V$ be an open C^1 , connected set. Show that for any $x \in U$,

$$\int_{\partial U} \langle \nabla_y G(x, y), \eta_U(y) \rangle d\sigma(y) = -1.$$

3. Consider the annulus $A = B(0, 2) \setminus B(0, 1) \subseteq \mathbb{R}^n$

- (1) [7 marks] Explicitly describe (with a formula for the pointwise value) a harmonic function $u \in C^2(\overline{A}, \mathbb{R})$ such that

$$u|_{\partial B(0,1)} \equiv 0, \quad \text{and} \quad u|_{\partial B(0,2)} \equiv 1.$$

- (2) [2 marks] Is u the unique harmonic function in $C^2(\overline{A}, \mathbb{R})$ satisfying the above boundary conditions? Explain.

4. [6 marks] Let $(u_n)_{n=1}^\infty \subset C^2(\overline{V}, \mathbb{R})$ be a sequence of functions satisfying

$$\Delta u_n = 0, u_n|_{\partial V} = g_n,$$

where $(g_n)_{n=1}^\infty$ is a sequence of continuous functions in ∂V satisfying $\|g_n\|_\infty \leq 1$. Show that $(u_n)_{n=1}^\infty$ has a subsequence which converges uniformly on compact sets to a harmonic function.

5. Let $\eta(y) = y/\|y\|$, for all $(0, 0) \neq y \in \mathbb{R}^2$. Consider the vector field $F : \mathbb{R}^2 \setminus \{(0, 0)\} \rightarrow \mathbb{R}^2$ defined by

$$F(x) = (|x| - 1) \sin(|x|) \eta(x).$$

- (a) [2 marks] Explain why a solution to the ODE exists locally for all initial points taken from $\mathbb{R}^2 \setminus \{(0, 0)\}$.
 (b) [2 marks] Explicitly describe the dependence of the solution at time t on the initial point obtained from Gronwall's inequality.
 (c) [2 marks] Describe the set of initial points for which the solution is constant in time.
 (d) [2 marks] Explicitly determine the maximal interval of existence of a solution of the IVP with initial value $x \in \mathbb{R}^2 \setminus \{(0, 0)\}$.
 (e) [2 marks] Determine the set of initial points for which the image of the curve which is the solution of the IVP is a compact set.